

One-shot Clustering for Policy Evaluation in Federated TD Learning with LFA

Bachelor's Thesis

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1 Introduction

The problem we are interested in solving is policy evaluation in the Federated RL setup [3]. Essentially, we aim to employ the TD(0) algorithm with linear function approximation and capitalize on the collaboration of agents belonging to the same cluster to rapidly converge to the Value function. To achieve this, we perform clustering of the agents after a sufficiently long time interval, such that they are correctly classified with high probability. We then exploit the speed-up due to the collaboration.

2 Preliminaries

A Markov Decision Process (MDP) is defined by the tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma), \quad (1)$$

where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, $P(s' | s, a)$ the transition probability from state s to s' under action a , $R(s, a)$ the immediate reward function, and $\gamma \in (0, 1)$ the discount factor. The goal of reinforcement learning is to determine the best action-selection strategy by defining a value function. For a given policy π , the state-value function is defined as

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right], \quad (2)$$

where the expectation is taken with respect to the trajectory induced by π .

However, when the state-action space is large, computing $V^\pi(s)$ exactly becomes infeasible. A common approach is to approximate the value function using a linear function approximator (LFA):

$$V^\pi(s; \theta) \approx \phi(s)^\top \theta, \quad (3)$$

where $\phi(s) \in \mathbb{R}^d$ is a feature vector associated with state s , and $\theta \in \mathbb{R}^d$ is a parameter vector to be learned. We collect all feature vectors into the tall matrix

$$\Phi = \begin{bmatrix} \phi(s_1)^\top \\ \phi(s_2)^\top \\ \vdots \\ \phi(s_{|\mathcal{S}|})^\top \end{bmatrix} \in \mathbb{R}^{|\mathcal{S}| \times d}. \quad (4)$$

Temporal-Difference (TD(0)) learning with linear function approximation updates the parameter vector θ iteratively using

$$\theta_{t+1} = \theta_t + \alpha_t \delta_t \phi(s_t), \quad (5)$$

where α_t is the step size, and δ_t is the TD error given by

$$\delta_t = R(s_t, a_t) + \gamma \phi(s_{t+1})^\top \theta_t - \phi(s_t)^\top \theta_t. \quad (6)$$

The TD(0) fixed-point equation can be written in terms of the projected Bellman operator as the solution to

$$A_i \theta_i^* = b_i, \quad (7)$$

where

$$A_i = \Phi^T D_i (\Phi - \gamma P_i \Phi), \quad (8)$$

$$b_i = \Phi^T D_i R_i \quad (9)$$

Thus, θ^* is the unique solution that minimizes the mean-squared projected Bellman error.

3 Problem Formulation

We have N agents, that belong to K distinct clusters, denoted as S_i^* , $i \in \{1, 2, \dots, K\}$, however we do not know these cluster identities. Note that we assume no heterogeneity within a cluster, all the agents within a cluster are exactly the same.

Let θ_i^* denote the optimal parameter associated with all agents in the K -th cluster, $A_i \theta_i^* = b_i$, and we are interested to see how the agents converge to this value.

4 Clustering in Federated TD Learning

To capture the heterogeneity among agents in different clusters, we impose the following assumptions [4], [2]

Assumption 1 (Markov Kernel Heterogeneity). *There exists a $\Delta P_{max} > 0$ and $\Delta P_{min} > 0$ such that for all agents with cluster identity $i, j \in [K]$, $i \neq j$, it holds that*

$$\Delta P_{min} \leq \|P^{(i)} - P^{(j)}\| \leq \Delta P_{max} \quad (10)$$

Assumption 2 (Reward Heterogeneity). *There exists a $\Delta R_{max} > 0$ and $\Delta R_{min} > 0$ such that for all $i, j \in [K]$, it holds that*

$$\Delta R_{min} \leq \|R^{(i)} - R^{(j)}\| \leq \Delta R_{max}. \quad (11)$$

Assumption 3 (Bounded Rewards). *There exists an $R_{max} > 0$ such that for all $i \in [K]$, it holds that*

$$R^{(i)} \leq R_{max} \quad (12)$$

Algorithm 1 Clustered Federated RL with One-Shot Clustering and Periodic Communication

- 1: **Input:** number of agents N , horizon T , pre-clustering steps T_1 , communication period τ , step size α
- 2: Initialize parameters $\theta_0^{(i)}$ for each agent $i \in [N]$

▷ **Phase I: Local updates before clustering**

- 3: **for** $t = 1, 2, \dots, T_1$ **do**
- 4: **for** each agent $i \in [N]$ in parallel **do**
- 5: Perform one local TD(0) update:

$$\theta_{t+1}^{(i)} = \theta_t^{(i)} + \alpha \delta_t^{(i)} \phi(s_t^{(i)}) \quad (13)$$

- 6: **end for**
- 7: **end for**

▷ **One-shot clustering at the server**

- 8: Agents send $\{\theta_{T_1}^{(i)}\}_{i=1}^N$ to the server
- 9: Server performs pair-wise clustering $\{S_s\}_{s=1}^K$ over agent models
- 10: Server sends each agent i its cluster identity $s(i)$

▷ **Phase II: Clustered training with periodic communication**

- 11: **for** $t = T_1 + 1, \dots, T$ **do**
- 12: **for** each agent $i \in [N]$ in parallel **do**
- 13: Perform one local TD(0) update
- 14: **end for**
- 15: **if** $t \bmod \tau = 0$ **then**
- 16: Each agent i sends local update $\theta_t^{(i)}$ to the server
- 17: **for** each cluster $S = 1, \dots, K$ **do**
- 18: Server computes average gradient direction within cluster:

$$g_s = \frac{1}{|S_s|} \sum_{i \in S_s} (\theta_t^{(i)} - \theta_{t-\tau}^{(i)}) \quad (14)$$

- 19: Update cluster parameter $\theta^{(s)}$:

$$\theta_{t+1}^{(s)} = \theta_t^{(s)} - \alpha g_s \quad (15)$$

- 20: **end for**
 - 21: Server broadcasts $\theta_{t+1}^{(s(i))}$ to each agent i according to its cluster identity $s(i)$
 - 22: **end if**
 - 23: **end for**
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5 Main Result and Analysis

5.1 Result

The convergence result we get is as follows, for each cluster [4]

$$\mathbb{E} \left[\|V_{\hat{\theta}_T} - V_{\theta^*}\|_D^2 \right] \leq \tilde{O} \left(\frac{\tau'^2 G^2 + \tau^2}{(T - T_1)^2} + \frac{c_{\text{quad}}(\tau') \nu^2}{N(T - T_1)} + \frac{c_{\text{lin}}(\tau') \tau \nu^4}{(T - T_1)^2} + Q(\epsilon, \epsilon_1) \right), \quad (16)$$

where

$$\tau' = \lceil \tau_{\text{mix}}(\alpha_T^2 \overline{D}) \rceil, \quad \alpha_T = \tau \alpha_\ell \alpha_g^{(T)}, \quad (17)$$

and $c_{\text{quad}}(\tau)$, $c_{\text{lin}}(\tau)$ are quadratic and linear functions in τ , respectively.

The term that we want to tackle is $Q(\epsilon, \epsilon_1)$, in the case of no clustering, this term never vanishes. But we would like that as the cluster identity estimates get better, which happens when larger number of initial steps are taken, this error term also decays to zero. Note that all the other terms decay to zero eventually.

Moreover, we define

$$Q(\epsilon, \epsilon_1) = \tilde{O}\left(\mathbb{E}[B(\epsilon, \epsilon_1)] \frac{G}{\nu} + \mathbb{E}[\Gamma^2(\epsilon, \epsilon_1)]\right) \quad (18)$$

$$B(\epsilon, \epsilon_1) = H\gamma \|\Delta P_{\max}\| \frac{|S_i \cap \bar{S}_i^*|}{|S_i|} + \|\Delta R_{\max}\| \frac{|S_i \cap \bar{S}_i^*|}{|S_i|} + H \frac{\|\Delta P_{\max}\|}{\min_j \|I - P^{(j)}\|} |S_1 \cap \bar{S}_1^*| \quad (19)$$

$$\Gamma(\epsilon, \epsilon_1) = \max_{i \in [K]} \frac{\kappa(\bar{A}_i)H}{1 - \kappa(\bar{A}_i) \frac{A(\epsilon)}{\delta_1}} \left(\frac{A(\epsilon)}{\delta_1} + \frac{b(\epsilon, \epsilon_1)}{\delta_2} \right) \quad (20)$$

$$A(\epsilon) = \gamma H \|\Delta P_{\max}\| \frac{|S_i \cap \bar{S}_i^*|}{|S_i|} + (1 + \gamma) \frac{\|\Delta P_{\max}\| |S_i \cap \bar{S}_i^*|}{\min_{j \in S_i} \|\mathbb{I} - P_j\|} \quad (21)$$

$$b(\epsilon, \epsilon_1) = \|\Delta R_{\max}\| \frac{|S_i \cap \bar{S}_i^*|}{|S_i|} + R_{\max} \frac{\|\Delta P_{\max}\| |S_i \cap \bar{S}_i^*|}{\min_{j \in S_i} \|\mathbb{I} - P_j\|} \quad (22)$$

where ν corresponds to a linear transform on the eigenvalue of the virtual MDP stationary matrix, H is the projection radius, $G = R_{\max} + 2H$. S_i are the agents classified into cluster i , post the clustering step. S_i^* denotes the true cluster identity of the agent.

5.2 Discussion

Note that if we take $T_1 = O(\sqrt{T})$, we can essentially ignore it in the terms $T - T_1$. Hence we achieve a linear speed-up in convergence, as can be seen in the second term of equation (16). Also, we notice that as the clustering error δ decreases, the term $Q(\epsilon, \epsilon_1)$, decays to 0, with δ . What is left to be seen, is how exactly δ varies with T_1 .

From the paper [1],

Theorem 1 (TD(0) Concentration Bound). *Let $\lambda \in (0, \min_{i \in [d]} \{\text{real}(\lambda_i(A))\})$, where $\lambda_i(A)$ is the i -th eigenvalue of A . Let $\alpha_n = (n + 1)^{-1}$. Then, under $\mathcal{A}_2$, for $\mu > 0$ and $\delta \in (0, 1)$, there exists a function $N(\mu, \delta)$*

$$N(\mu, \delta) = \tilde{O}\left(\max\left\{\left[\frac{1}{\mu}\right]^{1+\frac{1}{\lambda}} \left[\ln \frac{1}{\delta}\right]^{1+\frac{1}{\lambda}}, \left[\frac{1}{\mu}\right]^2 \left[\ln \frac{1}{\delta}\right]^3\right\}\right), \quad (23)$$

such that

$$\Pr\left\{\|\theta_n - \theta^*\| \leq \mu, \forall n \geq N(\mu, \delta)\right\} \geq 1 - \delta. \quad (24)$$

This results uses the step size $\alpha_n = \frac{1}{n+1}$, and banks on the following two assumptions

Assumption 4. *It holds that*

$$\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \leq K_s [1 + \|\theta_n - \theta^*\|^2], \quad (25)$$

where

$$M_{n+1} = r_n + \gamma \phi_n^\top \theta_n - \phi_n^\top \theta^* - [b - A\theta_n]. \quad (26)$$

Assumption 5. *All rewards and feature vectors are uniformly bounded. That is,*

$$\|\phi(s)\| \leq \frac{1}{2}, \quad \forall s \in \mathcal{S}, \quad \text{and} \quad |r(s, a, s')| \leq 1, \quad \forall s, s' \in \mathcal{S}, a \in \mathcal{A}. \quad (27)$$

We choose ϵ such that clustering happens correctly with probability $1 - \delta$. Specifically, we set

$$\mu = \frac{1}{2} \min_{\substack{i \neq j \\ i, j \in [K]}} \|\theta_i^* - \theta_j^*\|_2. \quad (28)$$

Let us consider the case when T_1 scales as $\sqrt{T}$. Also, for simplicity of analysis and writing, we consider the right term to be dominant in expression (23), we then have

$$\delta = O(e^{(-T^{1/6} \mu^{2/3})}) \quad (29)$$

Note that this means, that after T_1 steps, the probability of missclassification of a single agent is δ .

6 Conclusion

In this work, we rigorously establish guarantees for achieving linear speed-up through collaboration in federated reinforcement learning. By introducing a new model of heterogeneity, we analyze how the perturbation of the virtual MDP relative to the TD(0) fixed point governs learning efficiency, and we quantify the effect of erroneous clustering on this perturbation. Our results demonstrate that, under mild assumptions, collaboration remains provably beneficial even in the presence of imperfect agent grouping. An important direction for future work is to develop algorithms that alternate between estimating cluster identities and performing the federated update step.

7 Appendix

7.1 Justification of assumptions on cluster distance

$$\Phi\theta = V, \quad A\theta^* = b \quad (30)$$

$$A_i = \Phi^T D_i (\Phi - \gamma P_i \Phi), \quad b_i = \Phi^T D_i R_i \quad (31)$$

$$\|\theta_1^* - \theta_2^*\| = \|A_1^{-1} b_1 - A_2^{-1} b_2\| \quad (32)$$

$$= \|(\Phi^T D_1 (\Phi - \gamma P_1 \Phi))^{-1} \Phi^T D_1 R_1 - (\Phi^T D_2 (\Phi - \gamma P_2 \Phi))^{-1} \Phi^T D_2 R_2\| \quad (33)$$

$$= \|(\Phi - \gamma P_1 \Phi)^{-1} R_1 - (\Phi - \gamma P_2 \Phi)^{-1} R_2\| \quad (34)$$

$$= \|\Phi^{-1} (I - \gamma P_1)^{-1} R_1 - \Phi^{-1} (I - \gamma P_2)^{-1} R_2\| \quad (35)$$

$$= \|(I - \gamma P_1)^{-1} R_1 - (I - \gamma P_2)^{-1} R_2\| \quad (36)$$

$$= \|(I - \gamma P_1)^{-1} R_1 - (I - \gamma P_1)^{-1} R_2 + (I - \gamma P_1)^{-1} R_2 - (I - \gamma P_2)^{-1} R_2\| \quad (37)$$

$$= \|(I - \gamma P_1)^{-1} (R_1 - R_2) + R_2 ((I - \gamma P_1)^{-1} - (I - \gamma P_2)^{-1})\| \quad (38)$$

We are interested in finding upper and lower bounds based on the assumptions. We begin by finding the upper bound

$$\|\theta_1^* - \theta_2^*\| \leq \|(I - \gamma P_1)^{-1} (R_1 - R_2)\| + \|R_2 ((I - \gamma P_1)^{-1} - (I - \gamma P_2)^{-1})\| \quad (39)$$

$$\leq \frac{\Delta R_{max}}{(1 - \gamma)} + \frac{2R_{max}}{(1 - \gamma)} \quad (40)$$

We now work on the minimum bound

$$\|\theta_1^* - \theta_2^*\| \geq \|R_1 - R_2\| - \gamma \|P_2 R_2 - P_1 R_1\| \quad (41)$$

$$\geq \|\|R_1 - R_2\| - \gamma \|P_2 (R_2 - R_1) + (P_2 - P_1) R_1\|\| \quad (42)$$

7.2 Proof of $A(\epsilon)$ and $b(\epsilon, \epsilon_1)$

$$\|\bar{A}_i - \bar{A}\| = \|\Phi^T D^{(i)}(\Phi - \gamma P^{(i)}\Phi) - \Phi^T \bar{D}(\Phi - \gamma \bar{P}\Phi)\| \quad (43)$$

$$= \|\Phi^T D^{(i)}(\Phi - \gamma P^{(i)}\Phi) - \Phi^T D^{(i)}(\Phi - \gamma \bar{P}\Phi) + \Phi^T D^{(i)}(\Phi - \gamma \bar{P}\Phi)\Phi^T \bar{D}(\Phi - \gamma \bar{P}\Phi)\| \quad (44)$$

$$\leq \|\Phi^T D^{(i)}(\Phi - \gamma P^{(i)}\Phi) - \Phi^T D^{(i)}(\Phi - \gamma \bar{P}\Phi)\| \quad (45)$$

$$+ \|\Phi^T D^{(i)}(\Phi - \gamma \bar{P}\Phi) - \Phi^T \bar{D}(\Phi - \gamma \bar{P}\Phi)\| \quad (46)$$

$$\leq \gamma \|\Phi\|^2 \|D^{(i)}\| \|P^{(i)} - \bar{P}\| + \|\Phi\|^2 \|D^{(i)} - \bar{D}\| \|I - \gamma \bar{P}\| \quad (47)$$

$$\leq \gamma \|P^{(i)} - \bar{P}\| + (1 + \gamma) \|D^{(i)} - \bar{D}\| = A(\epsilon) \quad (48)$$

$$\|\bar{b}_i - \bar{b}\| = \|\Phi D^{(i)} R^{(i)} - \Phi \bar{D} \bar{R}\| \quad (49)$$

$$= \|\Phi D^{(i)} R^{(i)} - \Phi D^{(i)} \bar{R} + \Phi D^{(i)} \bar{R} - \Phi \bar{D} \bar{R}\| \quad (50)$$

$$\leq \|\Phi\| \|D^{(i)}\| \|R^{(i)} - \bar{R}\| + \|\Phi\| \|D^{(i)} - \bar{D}\| \|\bar{R}\| \quad (51)$$

$$\leq \|R^{(i)} - \bar{R}\| + R_{\max} \|D^{(i)} - \bar{D}\| = b(\epsilon, \epsilon_1) \quad (52)$$

Here we have used that $\|\Phi\| = 1$, $\|D^{(i)}\| = 1$, $\|\bar{D}\| = 1$.

Plugging this into the standard result,

$$\|\theta_i^* - \theta^*\| \leq \frac{K(A_i) \cdot H}{1 - K(A_i) \cdot \frac{A(\epsilon)}{\delta_1}} \left(\frac{A(\epsilon)}{\delta_1} + \frac{B(\epsilon, \epsilon_1)}{\delta_2} \right),$$

$$\delta_1 \leq \|A_i\|,$$

$$\delta_2 \leq \|\bar{b}_i\|.$$

7.3 Proof of $B(\epsilon, \epsilon_1)$

$$\begin{aligned}
\|\bar{g}(\theta) - \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \bar{g}_i(\theta)\| &= \left\| \Phi^T \bar{D}(T_\mu \Phi \theta - \Phi \theta) - \frac{1}{N} \sum_{i=1}^{|S_1|} \Phi^T D^{(i)}(T_\mu^{(i)} \Phi \theta - \Phi \theta) \right\| \\
&\leq \frac{1}{|S|} \sum_{i=1}^{|S|} \left\| \Phi^T \bar{D}(T_\mu \Phi \theta - \Phi \theta) - \Phi^T D^{(i)}(T_\mu^{(i)} \Phi \theta - \Phi \theta) \right\|, \\
&\text{using } \|\Phi\| = 1, \text{ and adding and subtracting terms} \\
&\leq \frac{1}{|S|} \sum_{i=1}^{|S|} \left\| \Phi^T \bar{D}(T_\mu \Phi \theta - \Phi \theta) - \Phi^T \bar{D}(T_\mu^{(i)} \Phi \theta - \Phi \theta) \right. \\
&\quad \left. + \Phi^T \bar{D}(T_\mu^{(i)} \Phi \theta - \Phi \theta) - \Phi^T D^{(i)}(T_\mu^{(i)} \Phi \theta - \Phi \theta) \right\| \\
&\leq \frac{1}{|S|} \sum_{i=1}^{|S|} \left(\left\| \Phi^T \bar{D}[(T_\mu - T_\mu^{(i)}) \Phi \theta] \right\| + \left\| \Phi^T (\bar{D} - D^{(i)})(T_\mu^{(i)} \Phi \theta - \Phi \theta) \right\| \right) \\
&\leq \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \left\| \bar{D} \left[\frac{1}{|S_1|} \sum_{j=1}^{|S_1|} R^{(j)} + \gamma \frac{1}{|S_1|} \sum_{j=1}^{|S_1|} P^{(j)} \Phi \theta - \Phi \theta \right] - \bar{D}(T_\mu^{(j)} \Phi \theta - \Phi \theta) \right\| \\
&\quad + \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \|\bar{D} - D^{(i)}\| \|T_\mu^{(i)} \Phi \theta - \Phi \theta\|, \\
&\leq \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \left\| \frac{1}{|S_1|} \sum_{j=1}^{|S_1|} R^{(j)} - R^{(i)} \right\| + \gamma \left\| \frac{1}{|S_1|} \sum_{j=1}^{|S_1|} P^{(j)} - P^{(i)} \right\| \|\Phi \theta\| \\
&\quad + \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \|\bar{D} - D^{(i)}\| \|T_\mu^{(i)} \Phi \theta - \Phi \theta\|.
\end{aligned}$$

Note that here $\bar{D}, \bar{g}(\theta)$ respectively correspond to the stationary probability distribution and steady state update direction of the virtual MDP of all the agents that belong to a particular cluster post the one-shot clustering step. We need to now carefully analyse these three terms that remain, and how they change as the clustering accuracy changes. We would like this to diminish as the clustering becomes better. Hence we have the first term as

$$\begin{aligned}
\frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \left\| \frac{1}{|S_1|} \sum_{j=1}^{|S_1|} R^{(j)} - R^{(i)} \right\| &\leq \frac{1}{|S_1|} \sum_{i=1}^{|S_1|} \left\| \frac{1}{|S_1|} |S_1 \cap S_1^*| (\Delta R_{max}) \right\|, \\
&= \frac{|S_1| \cap |S_1^*|}{|S_1|} \Delta R_{max}
\end{aligned}$$

Here $\Delta R_{max} = \sup_{i,j \in K} \|R^{(i)} - R^{(j)}\|$

The second term is bounded as

$$\gamma \left\| \frac{1}{|S_1|} \sum_{j=1}^{|S_1|} P^{(j)} - P^{(i)} \right\| \|\Phi \theta\| \leq \gamma H \frac{|S_1 \cap S_1^*|}{|S_1|} \|\Delta P_{max}\|$$

This follows simple since $\|\theta\| \leq H, \|\phi\| = 1$, and $\Delta P_{max} = \sup_{i,j \in K} \|P^{(i)} - P^{(j)}\|$.

Let us now proceed to bound the final term. We begin by analyzing $\|\bar{D} - D^{(i)}\|$. We know that $D = DP$, in general, as it is the stationary distribution of the markov chain corresponding to P .

$$\begin{aligned}
\bar{D} - D^{(i)} &= \frac{1}{|S_1|} \left(\bar{D} \sum_{j \in S_1} P^{(j)} - D^{(i)} \sum_{j \in S_1} P^{(j)} \right) \\
|S_1|(\bar{D} - D^{(i)}) &= |S_1 \cap S_1| (\bar{D} - D^{(i)})P^{(i)} + \bar{D} \sum_{j \in S_1 \cap \bar{S}_1^*} P^{(j)} - D^{(i)} \sum_{j \in S_1 \cap \bar{S}_1^*} P^{(j)} \\
&= |S_1 \cap S_1| (\bar{D} - D^{(i)})P^{(i)} + \bar{D} \sum_{j \in S_1 \cap \bar{S}_1^*} P^{(j)} - D^{(i)} \sum_{j \in S_1 \cap \bar{S}_1^*} P^{(j)} \\
&\quad + D^{(i)} \sum_{j \in S_1 \cap S_1^*} P^{(j)} - D^{(i)} \sum_{j \in S_1 \cap S_1^*} P^{(j)} \\
&\leq |S_1 \cap S_1^*| (\bar{D} - D^{(i)})P^{(i)} + (\bar{D} - D^{(i)}) \sum_{j \in S_i \cap \bar{S}_1} P^{(j)} + D^{(i)} |S_i \cap \bar{S}_1| \Delta P_{max} \\
|S_1 \cap S_1^*| (\bar{D} - D^{(i)}) (\mathbb{I} - P^{(i)}) + \sum_{j \in S_i \cap \bar{S}_1} (\bar{D} - D^{(i)}) (\mathbb{I} - P^{(j)}) &\leq D^{(i)} |S_i \cap \bar{S}_1| \Delta P_{max} \\
\|\bar{D} - D^{(i)}\| &\leq \frac{\|\Delta P_{max}\| |S_1 \cap \bar{S}_1^*|}{\min_{j \in S_1} \|\mathbb{I} - P_j\|}
\end{aligned}$$

7.4 Analyzing the fraction of misclassified agents

We are interested in understanding how $\frac{|S_1 \cap \bar{S}_1^*|}{|S_1|}$ and $|S_1 \cap \bar{S}_1^*|$ change with the clustering error. We would like the values to reduce as the error goes down, which in turn happens when the number of local TD(0) steps before clustering increases. Again, S_i denotes the cluster post the one-shot clustering step, S_i^* has the agents that have cluster identity as i , $\bar{S}_1^*$ has all the agents which don't belong to the i^{th} cluster.

Let the probability of misclassification be δ . Then,

$$\begin{aligned}
S_1 \cap \bar{S}_1^* &\subset \bar{S}_1^* \\
\mathbb{E}(|S_1 \cap \bar{S}_1^*|) &\leq \bar{S}_1^* \delta
\end{aligned}$$

Let us now bound the denominator term

$$|S_1| > \frac{1 - \delta}{2} |S_1^*|,$$

$$\text{with probability } \geq 1 - \exp\left(\frac{-2}{|S_1|} \left(|\bar{S}_1^*| \delta + |S_1^*| \frac{1 - \delta}{2}\right)^2\right).$$

We start by writing

$$\Pr\left(|S_1| > \frac{(1 - \delta)}{2} |S_1^*|\right) = \Pr\left(|S_1| - \mathbb{E}[|S_1|] > \frac{(1 - \delta)}{2} |S_1^*| - \mathbb{E}[|S_1|]\right).$$

Now, using the fact that the expected value of $|S_1|$ satisfies

$$\mathbb{E}[|S_1|] \leq |S_1^*|(1 - \delta) + |\bar{S}_1^*| \delta,$$

we have

$$\frac{(1 - \delta)}{2} |S_1^*| - \mathbb{E}[|S_1|] \geq -|S_1^*| \delta - |\bar{S}_1^*| \frac{(1 - \delta)}{2}.$$

Therefore,

$$\begin{aligned}\Pr\left(|S_1| > \frac{(1-\delta)}{2}|S_1^*|\right) &\geq \Pr\left(|S_1| - \mathbb{E}[|S_1|] > -|S_1^*|\delta - |\bar{S}_1^*|\frac{(1-\delta)}{2}\right) \\ &\geq 1 - \Pr\left(|S_1| - \mathbb{E}[|S_1|] \leq -|S_1^*|\delta - |\bar{S}_1^*|\frac{(1-\delta)}{2}\right).\end{aligned}$$

By Hoeffding's inequality, for independent bounded random variables,

$$\Pr(|S_1| - \mathbb{E}[|S_1|] \leq -t) \leq \exp\left(-\frac{2t^2}{|S_1|}\right).$$

Substituting $t = |\bar{S}_1^*|\delta + |S_1^*|\frac{(1-\delta)}{2}$, we obtain

$$\Pr\left(|S_1| > \frac{(1-\delta)}{2}|S_1^*|\right) \geq 1 - \exp\left(\frac{-2}{|S_1|} \left(|\bar{S}_1^*|\delta + |S_1^*|\frac{(1-\delta)}{2}\right)^2\right).$$

This completes the proof, and gives us the required high probability bound.

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